

Article

Real-Time Water Quality Monitoring System in an Aquaponics Pilot Culture

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Abstract

Water-quality monitoring is critical for maintaining the symbiotic balance and productivity of aquaponic systems. This study presents the design, implementation, and evaluation of a remote, real-time monitoring system based on the Internet of Things (IoT) paradigm. The system continuously monitors the key parameters of temperature, pH, electrical conductivity, total dissolved solids, salinity, dissolved oxygen, turbidity, and total suspended solids. Utilizing a modular architecture, the platform provides real-time visualization, cloud-based data management, and automated alerts via SMS and e-mail to notify operators of deviations from established tolerance ranges. The system was experimentally validated over a six-month period in a pilot-scale aquaponics system cultivating common carp (*Cyprinus carpio*). Statistical analysis demonstrated a 97% data acquisition reliability rate. Furthermore, no statistically significant differences ($p > 0.05$) were observed between the sensor-based measurements and reference laboratory analyses, confirming the system's high accuracy. This versatile and cost-effective tool enables data-driven decision-making, facilitates timely interventions to reduce production losses, and ensures the long-term environmental stability of integrated aquaculture systems.

Keywords: sensors; water quality; real-time monitoring; aquaponics; IoT



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1. Introduction

Aquaponics is defined as the integrated co-cultivation of aquatic organisms and plants, in which effluent from aquaculture serves as a nutrient source for plant growth, while plants, through nutrient uptake, improve water quality for reuse in aquaculture or for final discharge [1]. This system consists of the coupling of a recirculating aquaculture system with a hydroponic production unit operating simultaneously [2]. Research on aquaponics has expanded significantly since 2010, with the primary objective of improving water quality and reducing excess ammonia accumulation in aquaculture systems. Plants function mainly as biofilters [3], as poor water quality directly affects fish growth rate, feed intake, health status, and overall productivity [4].

Aquaculture in Mexico has reached varying levels of development, ranging from experimental production, as in the case of white fish, native mojarra, abalone, scallops,

mussels, lobster, and snails, to large-scale commercial production of species such as catfish, carp, tilapia, trout, oysters, shrimp, and freshwater prawns. Current national programs promote the integration of fish farming as a complementary activity to agriculture, aiming to improve access to high-quality protein for human consumption. In 2023, the National Commission of Aquaculture and Fisheries [5] reported national fisheries and aquaculture production exceeding 1.9 million tons. Of this total, more than 1.74 million tons corresponded to capture fisheries, while aquaculture production exceeded 237,250 tons during the same year [5,6].

The State of Mexico ranks first nationally in aquaculture production among the country's landlocked states. In recent decades, several species have been cultivated, with Mexican rainbow trout standing out as one of the most important, particularly at the Calimaya Aquaculture Center and in the municipalities of Jocotitlán, Ocoyoacac, and Valle de Bravo. Additionally, the state contains more than 21,000 hectares of flooded inland waters, representing approximately 1.2% of the national total, along with 11,859 reservoirs ranging from 1 to 10 hectares in area. In 2020, aquaculture production in these water bodies reached 24,425 tons [7,8].

The design and operation of aquaponic systems represent a major technological challenge when high yields and product quality are desired for both fish and plants. Because aquaponics functions as a controlled and symbiotic environment, numerous parameters must be continuously regulated, including light intensity, temperature, pH, humidity, dissolved oxygen, and electrical conductivity. This complexity makes the measurement, analysis, and management of water quality parameters particularly demanding. With the advancement of automation, smart agricultural strategies, and digital connectivity, online monitoring has become a key tool for improving the performance of aquaponic systems [2,9]. Expected benefits of intelligent automation include a substantial reduction in manual labor, enhanced process control through improved accessibility and connectivity, and the use of computational tools to enable data-driven decision-making [10].

The expected benefits of intelligent automation are a significant reduction in manual work, better control of processes by increasing the accessibility and connectivity of parameters, and the use of computing capabilities to make data-driven decisions [10]. The classic method of water quality monitoring consists of a qualified person taking measurements of a variable with a multiparametric instrument. When an alarm is detected due to abnormal levels of some parameter, a water sample is taken and transported to the chemical laboratory to analyze the substances contained in the water [11]. Monitoring by analyzing samples in the laboratory is a time-consuming and complicated process that often does not provide the required information or the large data sets necessary to describe the temporal evolution of the processes of interest [12].

Real-time monitoring systems are widely used to assess water quality in freshwater systems, including aquaculture and fish farming operations [13], surface waters [14], and groundwater [15]. Continuous water-quality monitoring enables timely operational decisions and corrective actions to maintain optimal nutrient conditions for organism development, ensuring aquaculture and aquaponic systems operate efficiently and sustainably at larger scales [16]. These systems also enable continuous verification of parameter stability and sensor performance. A typical monitoring system architecture consists of three subsystems responsible for data acquisition, control, and communication, along with data management [12]. These subsystems may include platforms [17], modules [18,19], and sensor nodes [20].

Modern aquaponic monitoring systems increasingly adopt the Internet of Things (IoT) paradigm to enhance data acquisition, transmission, and analysis. Several systems employ sensors to continuously measure parameters such as temperature, pH, and dissolved oxy-

gen. Data are transmitted using microcontrollers such as NodeMCU ESP32 and managed through mobile applications, enabling real-time alerts under critical conditions [21]. Other platforms integrate multiple sensors and wireless communication technologies such as NB-IoT and LoRaWAN for long-distance transmission and real-time visualization [22]. Automated monitoring systems have also incorporated smart feeding technologies combined with environmental sensing, allowing real-time notifications when operating parameters exceed established thresholds, thereby improving the system's responsiveness to disturbances [21].

In this study, a remote water-quality monitoring system based on a coordinating node connected to both a private local network and a public TCP/IP and SMS communication network was designed, implemented, and evaluated in a pilot-scale aquaponics system. Using multiparameter sensors, the system provides real-time information on water quality at the main culture tank as a critical single control point to verify that operating conditions remain within the tolerance range required for each organism. The developed system was experimentally evaluated through continuous monitoring of key water-quality parameters, including temperature, turbidity, and salinity, thereby validating its applicability under real operating conditions and its contribution to quality control in aquaponic fish production.

The collected data can be accessed and analyzed remotely via the Internet at any time to assess system status and detect operational changes. In addition, the system generates alarms within seconds in response to anomalous environmental fluctuations. Long-term data acquisition allows detailed characterization of the temporal dynamics of water-quality conditions, enabling real-time verification and early intervention to reduce production losses and improve overall system stability.

The main novelty of this platform lies in the orchestration of an extensive 8-parameter sensing framework paired with a subscription-free cloud data architecture, delivering high-frequency operational security without recurring software overhead costs during long-term aquaponic operations.

2. Materials and Methods

2.1. System Design

The water-quality monitoring system developed in this work was designed to continuously acquire and display, in real time, eight key parameters relevant to aquaponic operation. During the monitoring experiments, four multiparameter water-quality sensors were deployed in the aquaponic system. These probes measured temperature, pH, electrical conductivity (EC), total dissolved solids (TDS), salinity, dissolved oxygen (DO), turbidity and total suspended solids. All sensors were factory-calibrated prior to installation and subsequently programmed and integrated into the aquaponic system.

To validate the accuracy of the monitoring system, field measurements were compared against reference values obtained using laboratory instruments previously calibrated with certified standards. After individual programming and testing of each sensor, the corresponding codes were integrated into a single acquisition routine. The complete system was then tested under operating conditions until stable and consistent responses were obtained, and standard solutions for each parameter were measured to confirm correct performance.

The monitoring architecture consists of three main nodes: (i) a sensing node, (ii) a coordinating node, and (iii) a visualization node. Together, these subsystems enable basic self-maintenance and self-diagnosis under unsupervised operating conditions. Using dedicated data acquisition and monitoring software, the coordinating node communicates with each remote probe via the Modbus protocol, sending commands and receiving the corresponding measurements. The collected data are then forwarded to a monitoring terminal, where they are received via e-mail and simultaneously stored on a local memory

card for subsequent uploading and analysis. A schematic representation of the monitoring system is shown in Figure 1.

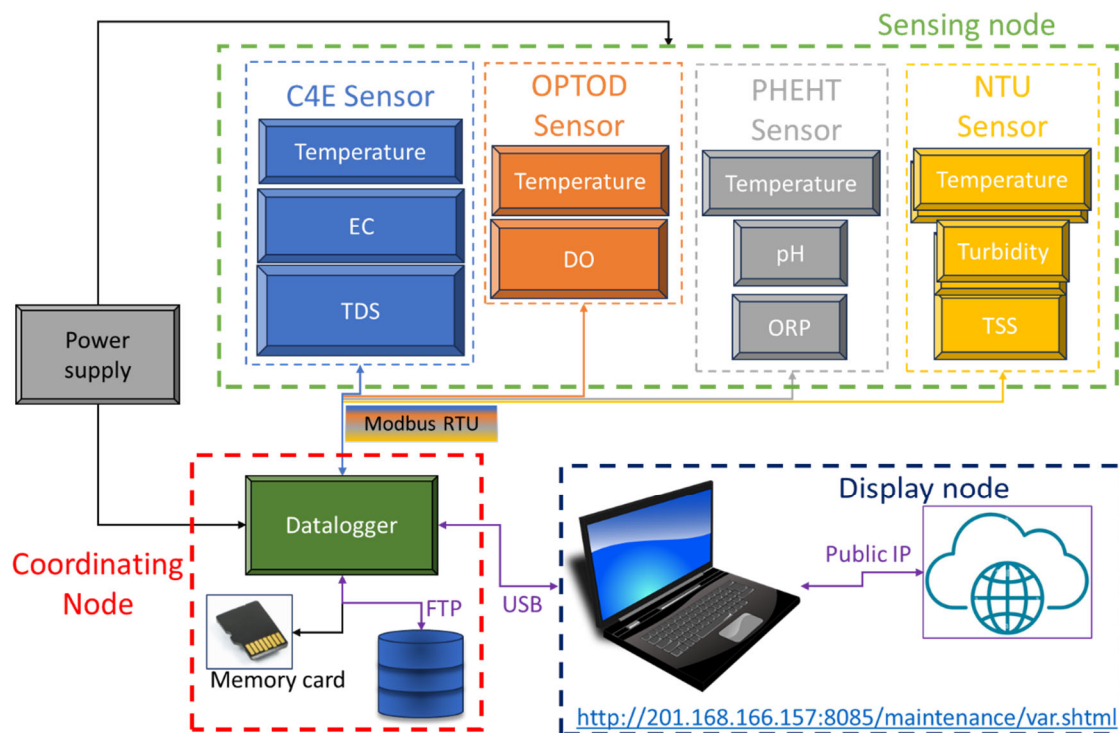


Figure 1. General scheme of the real-time water quality monitoring system.

The IoT module was programmed to automatically read the eight water-quality parameters from the fish tank of the aquaponic system at 15-min intervals. The system was also configured to generate alarms via SMS and e-mail whenever any measured parameter exceeded its predefined acceptable range, thereby notifying the operator of abnormal fluctuations. In addition, the system periodically (every 12 h) uploads recorded measurement data to a cloud storage service (Microsoft OneDrive; user-defined folder), allowing remote access, long-term storage, and further analysis of the temporal evolution of water quality.

2.2. Experimental Setup

Real-time monitoring of the water-quality parameters in the aquaponics system was initiated after a stabilization and adaptation period of three months. Subsequently, the eight selected water-quality parameters were continuously monitored over a six-month experimental period. In parallel, grab water samples were collected twice per week and analyzed using conventional laboratory methods. These measurements were used to validate and verify the accuracy of the real-time sensor data through direct comparison.

Physicochemical parameters play a fundamental role in ensuring the sustainability and productivity of aquaculture systems. The experimental setup consisted of a pilot-scale aquaponics system composed of the following interconnected units: (1) a $100 \times 100 \times 100$ cm plastic culture tank with a total working volume of 1000 L for common carp (*Cyprinus carpio*) cultivation, stocked with approximately 500 individuals; (2) a peristaltic pump used to circulate the water; (3) a hydroponic cultivation unit for strawberry production; (4) a secondary tank containing the aquatic macrophyte *Lemna gibba*; (5) a constructed wetland planted with *Cyperus articulatus* (tule), which functioned as a biological filtration unit; and (6) a water-storage tank from which the treated water was manually returned to the fish-culture tank to complete the recirculation loop.

The sensors used to monitor the different water-quality parameters were fully submerged in the fish-culture tank and operated continuously over the six-month monitoring period to evaluate the temporal behavior and stability of water quality at this critical control point of the system (Figure 2).

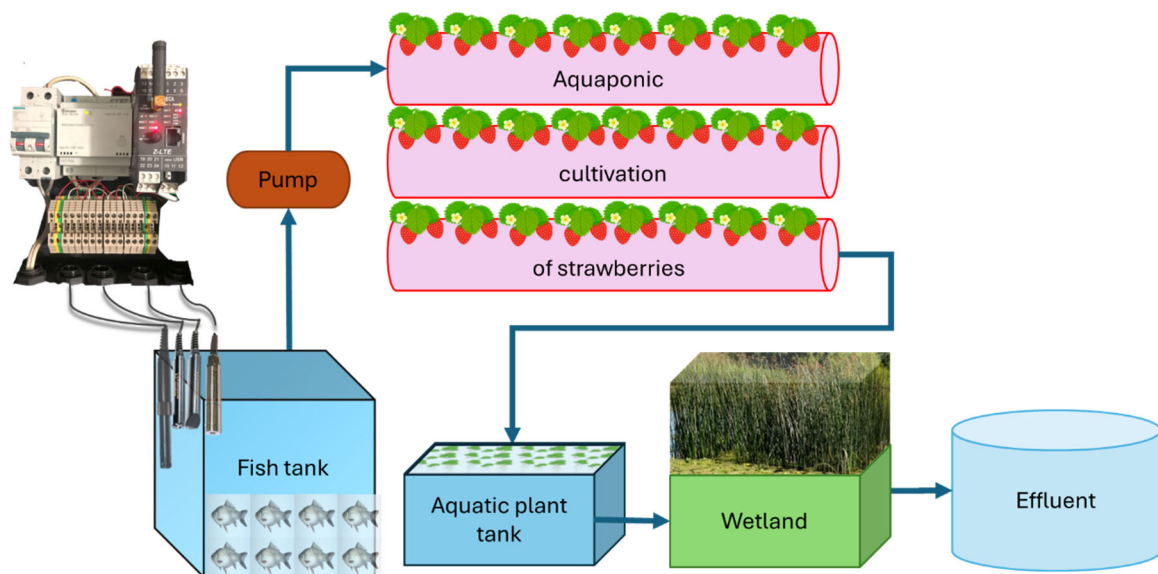


Figure 2. Diagram of the monitored aquaponics system.

2.3. Sensing Node

The sensors collected and acquired data from eight water-quality parameters to assess the environmental conditions required for optimal fish and plant development in the aquaponics system. Temperature and dissolved oxygen (DO) were continuously monitored to ensure that they remained within the acceptable ranges for aquaculture, which for common carp (*Cyprinus carpio*) are 10–28 °C and 2–6 mg/L, respectively [16,23].

An Aqualabo (Champigny, France) C4E potentiometric sensor equipped with four electrodes, a polyurethane protective cover, and a PVC body was used to monitor electrical conductivity (EC), salinity, temperature, and total dissolved solids (TDS). The reported sensitivity of the EC measurement ranges from 0.01 to 1 $\mu\text{S}/\text{cm}$. Conductivity was measured in real time ($\mu\text{S}/\text{cm}$), and the data were transmitted to a data logger for storage on a memory card. Prior to the monitoring campaign, the correct operation of the sensor was verified using certified standard solutions, as described previously. For common carp, a recommended range of 200–1800 $\mu\text{S}/\text{cm}$ has been reported [23]. Accordingly, alarm thresholds were programmed to activate when conductivity exceeded 1500 $\mu\text{S}/\text{cm}$, allowing early operator intervention before reaching the maximum permissible limit.

An Aqualabo PHEHT combined sensor was used to measure pH, oxidation–reduction potential (ORP), and temperature. The operational ranges of this sensor are pH 0–14, redox potential –1000 to +1000 mV, and temperature –10 to +50 °C. The sensing electrode is manufactured from special glass with an Ag/AgCl reference electrode and KCl electrolyte. The reported resolution for all three parameters is 0.01 units. Prior to deployment in the aquaponics system, pH calibration was performed using standard buffer solutions of pH 4.0, 7.0, and 10.0, yielding deviations of less than 5%. This sensor was used to monitor acidity/alkalinity in the fish tank and verify compliance with the optimal pH range 6.5–8.2 for common carp [23]. Alarm notifications were programmed to be triggered when pH values fell outside this acceptable range.

Dissolved oxygen is characterized by relatively low solubility and availability in aquatic environments compared with terrestrial systems and plays a critical role in funda-

mental biochemical processes such as oxidation, nitrification, and organic matter decomposition [4]. Dissolved oxygen was measured using an Aqualabo OPTOD optical sensor based on luminescence quenching technology, in accordance with ASTM Standard D888-05. This sensor provides a resolution of 0.01 mg/L and operates within a measurement range of 0–20 mg/L. Sensor performance was verified using a zero-oxygen solution [24], which was independently measured with a Hach HQ440d multioximeter (Loveland, CO, USA), yielding a DO value of zero.

Turbidity, defined as the loss of water transparency caused by suspended particles, was measured using an Aqualabo NTU nephelometric turbidity sensor. This instrument has a measuring range of 0–4000 NTU (or 0–4500 mg/L) with a resolution between 0.01 and 1 NTU. The sensor incorporates components made of PVC, DELRIN, quartz, polyurethane, and polyamide polymer resins. As in the previous cases, calibration was verified using a formazin standard in accordance with the Mexican standard NMX-AA-038-SCFI-2001 [25]. The standard was also analyzed using a Hach DR/890 colorimeter (Loveland, CO, USA) to confirm measurement accuracy.

2.4. Coordinating Node

A Z-LTE-WW data logger was used for sensor data acquisition via the MODBUS communication protocol, and system programming was performed using the SeAL Graph Editor software v2.7.7. The acquired data are processed locally, stored on a memory card, transmitted via e-mail, and uploaded to cloud storage. When the controller detects measurements outside the predefined operating limits, an alarm is automatically generated. The operational logic of the system was designed using a modular software framework, which provides a high degree of flexibility. This modular architecture enables individual components to be independently assembled, modified, added, or removed without compromising the functionality of the remaining system elements [19,26].

The proposed software architecture is composed of three fundamental functional layers: Connectivity:

The real-time water-quality monitoring system uses the SMTP (Simple Mail Transfer Protocol) IoT communication protocol to transmit data to the cloud. SMTP enables the automatic sending of data files via electronic mail, allowing real-time access and visualization on a personal computer or equivalent device.

Security:

Two-factor authentication was implemented for the assigned email account to ensure secure data transmission. An application-specific password, generated by the email service provider, was used to authenticate access for external applications and automated services. Data files are automatically transferred to the cloud through a secured application interface.

Scalability:

All collected data are stored in the cloud, where they can be accessed and analyzed at any time [9]. The cloud-based data management environment allows dynamic expansion of storage capacity as required, ensuring long-term operability and adaptability of the monitoring system.

2.5. Display Node

To enable real-time monitoring of the IoT system through a web-based interface, a standard web browser was used on a personal computer, smartphone, or tablet, allowing real-time visualization of the variables recorded by the data logger. Through this interface, users can monitor device status, detect system errors, and download data log files stored on the memory card. An Internet connection is established to enable communication using the Hypertext Transfer Protocol (HTTP), which is widely adopted for data transfer in IoT-based

systems. HTTP was selected due to its reliability in maintaining stable communication links and its support for secure encrypted data transmission, ensuring efficient and protected data exchange between devices [19].

The IoT platform database was implemented to visualize all monitored parameters through a dashboard-style website. This interface was designed to minimize manual workload and facilitate rapid interpretation and analysis of sensor data by end-users. Power Automate applications were employed to automatically transfer e-mails generated by the data logger to cloud storage on Microsoft OneDrive. Subsequently, data analysis and graphical visualization were performed using spreadsheet-based processing tools.

2.6. Measurement of Water Quality Parameters in the Laboratory

Grab water samples were collected twice per week at a depth of 30 cm in the fish tank during the same period in which the automatic monitoring system was operating. Sampling was consistently performed at approximately 11:00 a.m. Each sample was transferred to a 1 L polyethylene container and immediately transported to the laboratory for physicochemical analysis.

Electrical conductivity, salinity, and total dissolved solids (TDS) were determined using a Thermo Scientific Orion Star Series A329 multiparameter (Waltham, MA, USA) meter equipped with an Orion 013010MD probe. The instrument was pre-calibrated using two certified conductivity standards (1410 $\mu\text{S}/\text{cm}$ and 12,900 $\mu\text{S}/\text{cm}$). pH measurements were performed using the same multiparameter meter fitted with an Orion 8107UWMMMD probe, which was calibrated with standard buffer solutions at pH 4.00, 7.00, and 10.00 prior to analysis. Measurements were recorded using the stability indicator displayed on the instrument and under selectable reading modes corresponding to conductivity, salinity, and TDS.

Turbidity and total suspended solids (TSS) were quantified using a Hach DR/890 colorimeter (Loveland, CO, USA) following the absorptiometric method. Calibration curves provided by the manufacturer were employed for turbidity quantification, and quality control was verified using a 200 NTU formazin standard solution. The turbidity determination is based on the scattering and absorption of light by suspended particles in the water sample. The measured turbidity is influenced by particle size, shape, color, and refractive index; therefore, deionized water was used as a reference blank to compensate for any background absorbance contributed by the reagents.

Dissolved oxygen (DO) was measured using a 300 mL Wheaton Winkler bottle (Montgomery, AL, USA) for sample collection, followed immediately by analysis with a Hach HQ440d multiparameter meter coupled to an LDO101 optical probe. The probe was calibrated using a zero-oxygen solution prior to measurements. The optical method operates within a measurement range of 0.1–20.0 mg/L, with a saturation range of 1–200% and a resolution of 0.1 mg/L. All physicochemical measurements were performed in triplicate to ensure analytical reliability.

To evaluate the operational accuracy and reliability of the developed real-time monitoring system, the online sensor datasets were rigorously validated against the parallel laboratory analytical measurements ($n = 30$ per group). Given the paired nature of the experimental design, where water samples for laboratory analysis were collected and compared simultaneously with the continuous sensor logs twice a week over a six-month period, a One-Way Repeated Measures Analysis of Variance (RM-ANOVA) was performed for each water quality parameter. This statistical approach effectively controls for temporal and seasonal variability across the monitoring timeline, focusing the analysis strictly on the variance attributed to the measurement methods (IoT system vs. laboratory standard). Assumptions of normality and sphericity were verified using the Shapiro–Wilk and Mauchly's

tests, respectively. Statistical significance was evaluated at a threshold of $\alpha = 0.05$. Furthermore, to assess the practical magnitude of the differences independently of the sample size constraints, the effect size for each parameter was determined using partial Eta-squared (η_p^2). All statistical computations were executed using Minitab statistical software v21.

3. Results and Discussion

3.1. Communication and Data Acquisition

The entire monitoring system was tested and verified over a six-month period, from June to December 2024. The normality (Shapiro–Wilk test) and homoscedasticity (Levene’s test) of the dataset were confirmed ($p > 0.05$); therefore, the results are presented as mean $\pm$ standard deviation (SD). Statistical analysis of the data collected from all monitoring nodes indicates that the system operated reliably, with a data acquisition reliability rate of 97%, calculated as the ratio of the 17,050 successfully logged data points against the theoretical 17,568 expected readings generated by the 15-min sampling interval over the 183-day campaign. This reliability exceeds the 95.2% data availability reported by [17] for comparable monitoring systems under similar environments.

3.2. Verification of Online Measurements of Water Quality Parameter

To verify the proper operation of the monitoring system and the accuracy of the measurements, grab samples were collected from the fish tank twice per week during the monitoring period. The following parameters were measured in these samples using laboratory instrumentation: temperature, pH, dissolved oxygen, turbidity, total suspended solids (TSS), salinity, electrical conductivity, and total dissolved solids (TDS) [27]. All laboratory instruments were calibrated prior to analysis using appropriate calibration standards in accordance with the manufacturers’ operating manuals.

Verification of the monitoring system measurements is essential to ensure their reliability and accuracy. In this study, the data obtained through conventional analytical techniques were compared with those recorded by the sensors of the real-time monitoring system in order to evaluate the quality and consistency of the measurements for the parameters described above.

Table 1 presents a comparison of the mean values obtained for selected water-quality parameters measured both by the online monitoring system and by laboratory instrumentation over the six-month monitoring period.

Table 1. Comparison of the mean values obtained for some parameters measured in the water by the online monitoring system and the laboratory equipment.

	Temperature (°C)		Electrical Conductivity (µS/cm)		Salinity (‰)		Total Dissolved Solids (mg/L)		Dissolved Oxygen (mg/L)		pH		Turbidity (NTU)		Total Suspended Solids (mg/L)	
	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2
Mean	15.16	15.56	922	923	0.494	0.495	465	467	7.1	7.4	8.5	8.7	31	38	46	49
Median	15.03	15.33	918	924	0.492	0.496	463	465	6.7	6.6	8.5	8.6	28	34	42	44
Standard Deviation	1.25	1.14	23	27	0.012	0.013	12	17	2.7	3	0.3	0.4	16	15	24	23
Minimum data	12.50	13.40	886	866	0.475	0.473	447	442	2.5	2.6	7.8	7.9	8	11	12	13
Maximum data	16.88	17.18	969	975	0.520	0.520	489	527	15.5	15.9	9.1	9.4	61	67	96	100
Coefficient of variation	8.3	7.3	2.5	3	2.5	2.7	2.5	3.6	38	41	3.5	4.6	50	48	51	50

1. Measurement with the sensor of the real-time monitoring system; 2. Measurement with the laboratory equipment; $n = 30$.

The measurements obtained for the eight monitored parameters showed overall high accuracy. The observed standard deviation ranges were 1.14–1.25 for temperature, 23–27 for electrical conductivity, 0.012–0.013 for salinity, 12–17 for total dissolved solids (TDS),

2.7–3.0 for dissolved oxygen (DO), 0.3–0.4 for pH, 15–16 for turbidity, and 23–24 for total suspended solids (TSS).

A One-Way Repeated Measures ANOVA (RM-ANOVA) was executed to compare the online sensor datasets against the parallel reference laboratory analyses ($n = 30$) in order to rigorously verify the operational accuracy of the continuous monitoring system. The statistical outputs confirmed that the real-time monitoring system operates with high analytical fidelity, showing no statistically significant differences between the two measurement methods across all evaluated physicochemical parameters ($p > 0.05$). Specifically, the temperature showed high equivalence between the sensor and laboratory methods, yielding an F-statistic of $F(1, 29) = 32.89$ with a p -value of 0.214, paired with a negligible effect size ($\eta_p^2 = 0.012$). The ionic and dissolved parameters demonstrated near-identical distributions, as evidenced by the statistical parameters of electrical conductivity ($F(1, 29) = 0.01$, $p = 0.927$, $\eta_p^2 < 0.001$), salinity ($F(1, 29) = 0.15$, $p = 0.702$, $\eta_p^2 = 0.005$), and total dissolved solids ($F(1, 29) = 0.36$, $p = 0.554$, $\eta_p^2 = 0.012$). Dissolved oxygen shows no a significance variance ($F(1, 29) = 0.98$, $p = 0.331$, $\eta_p^2 = 0.033$). Notably, pH remained statistically equivalent between both monitoring frameworks ($F(1, 29) = 4.21$, $p = 0.051$, $\eta_p^2 = 0.126$), successfully positioning its exact probability value above the critical significance threshold. Finally, despite the inherent environmental fluctuations driven by episodic feeding events, the physical parameters of turbidity ($F(1, 29) = 3.32$, $p = 0.079$, $\eta_p^2 = 0.103$) and total suspended solids ($F(1, 29) = 0.35$, $p = 0.559$, $\eta_p^2 = 0.012$) mathematically confirmed the analytical equivalence of the IoT optical sensors. Collectively, the consistently low magnitude of the partial Eta-squared values ($\eta_p^2 \leq 0.126$) demonstrates that the variance between the automated logs and the benchmark laboratory equipment is practically non-significant, reinforcing the suitability of the remote platform for precision aquaculture management.

Total suspended solids (TSS) and turbidity were the parameters that exhibited the greatest variability, with coefficients of variation (CV) of 50–51% and 48–50%, respectively. This indicates substantial fluctuations in these parameters in the fish-tank water quality during the monitoring period. It is important to note that the highest TSS and turbidity values were observed during fish-feeding events, which likely increased the concentration of suspended particles in the tank water [21].

These parameters were followed by temperature, which showed a CV of 7.3–8.3%, and pH, with a CV of 3.5–4.6%. Lower variability was observed for total dissolved solids (TDS), electrical conductivity, and salinity, with CV ranges of 2.5–3.6%, 2.5–3.0%, and 2.5–2.7%, respectively. The relatively low CV values for these latter parameters indicate that their concentration remained comparatively stable in the aquaculture system during the monitoring period.

The observed stability of pH reflects the effective performance of the biofiltration process occurring in the constructed wetlands planted with *Lemna gibba* and *Cyperus articulatus*. This observation is consistent with previous studies reporting the buffering capacity and phytoremediation efficiency of aquatic macrophytes in integrated aquaculture systems [1]. Notably, most monitored parameters remained within the conventional tolerance limits reported for *Cyprinus carpio* [23], except for episodic dissolved oxygen (DO) deviations driven by manual aeration management. These findings confirm that while the real-time monitoring system effectively flags operational fluctuations, fully automated control loops are required to optimize production and biological stability.

As an example, Figure 3 presents the comparison between the dissolved oxygen (DO) data obtained from the real-time monitoring system and those measured using laboratory instrumentation. A close agreement and similar trend between the two datasets can be observed. The monitoring system achieved 97% valid data acquisition, which exceeds the 95.2% reliability reported by [17] for comparable monitoring systems. Furthermore,

the DO measurement accuracy obtained in this study (± 2.7 mg/L) is consistent with the performance of IoT-based aquaculture sensors reported in recent studies [19].

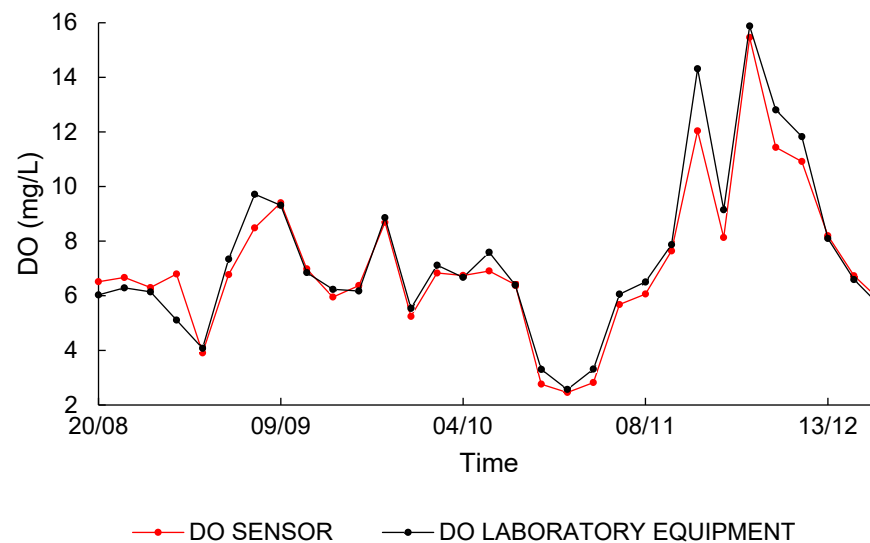


Figure 3. Comparison between the data obtained by the DO sensor of the real-time monitoring system and those obtained with the laboratory equipment.

The average DO concentration measured by the monitoring system was 7.1 ± 2.7 mg/L, while laboratory measurements yielded an average value of 7.4 ± 3.0 mg/L (Table 1). A strong correlation was observed between the two measurement methods ($R^2 = 0.94$, $p < 0.05$). In addition, analysis of variance indicated no statistically significant differences between the sensor measurements and the laboratory measurements at the 95% confidence level ($p > 0.05$). These results demonstrate that the real-time monitoring system provides reliable DO measurements with a minimal margin of error.

It is important to note that real-time measurements obtained using submerged sensors may be affected by partial obstruction or fouling of the sensing surface, a limitation that does not occur when samples are analyzed under controlled laboratory conditions. Conversely, laboratory samples may experience changes in dissolved oxygen concentration between the time of collection and the moment of analysis due to biological activity or gas exchange with the atmosphere.

Minor deviations in dissolved oxygen measurements at low concentrations (< 3 mg/L) have been attributed to the accumulation of biofilms on optical sensors during prolonged operation [17]. In the present study, the weekly maintenance protocol implemented for the real-time monitoring system significantly reduced these deviations. Specifically, measurement variability was reduced by approximately 78% compared with monitoring systems that operate without regular maintenance protocols [26]. These results support the operational reliability of the real-time monitoring system for long-term water-quality monitoring applications.

The ability of the real-time monitoring system to measure water-quality parameters accurately, precisely, and continuously increases confidence in the obtained results. By minimizing the need for human intervention, the system improves operational efficiency by reducing the time and effort required for data collection. The comparison between the measurements obtained from the real-time monitoring system and those obtained through laboratory analyses confirms that the proposed system can monitor water-quality parameters with high reliability. Therefore, the system represents a valuable tool for water-quality management in aquaponics systems, provided that appropriate operating and maintenance conditions are maintained.

3.3. Hardware

The water-quality monitoring system was successfully installed and operated in the fish tank of the aquaponics system, as shown in Figure 4. Figure 4a shows the four probes used to measure the water-quality parameters in the fish tank. These probes were installed approximately 30 cm from the inflow and outflow points to avoid waste accumulation and to optimize measurement accuracy, as recommended by [28].

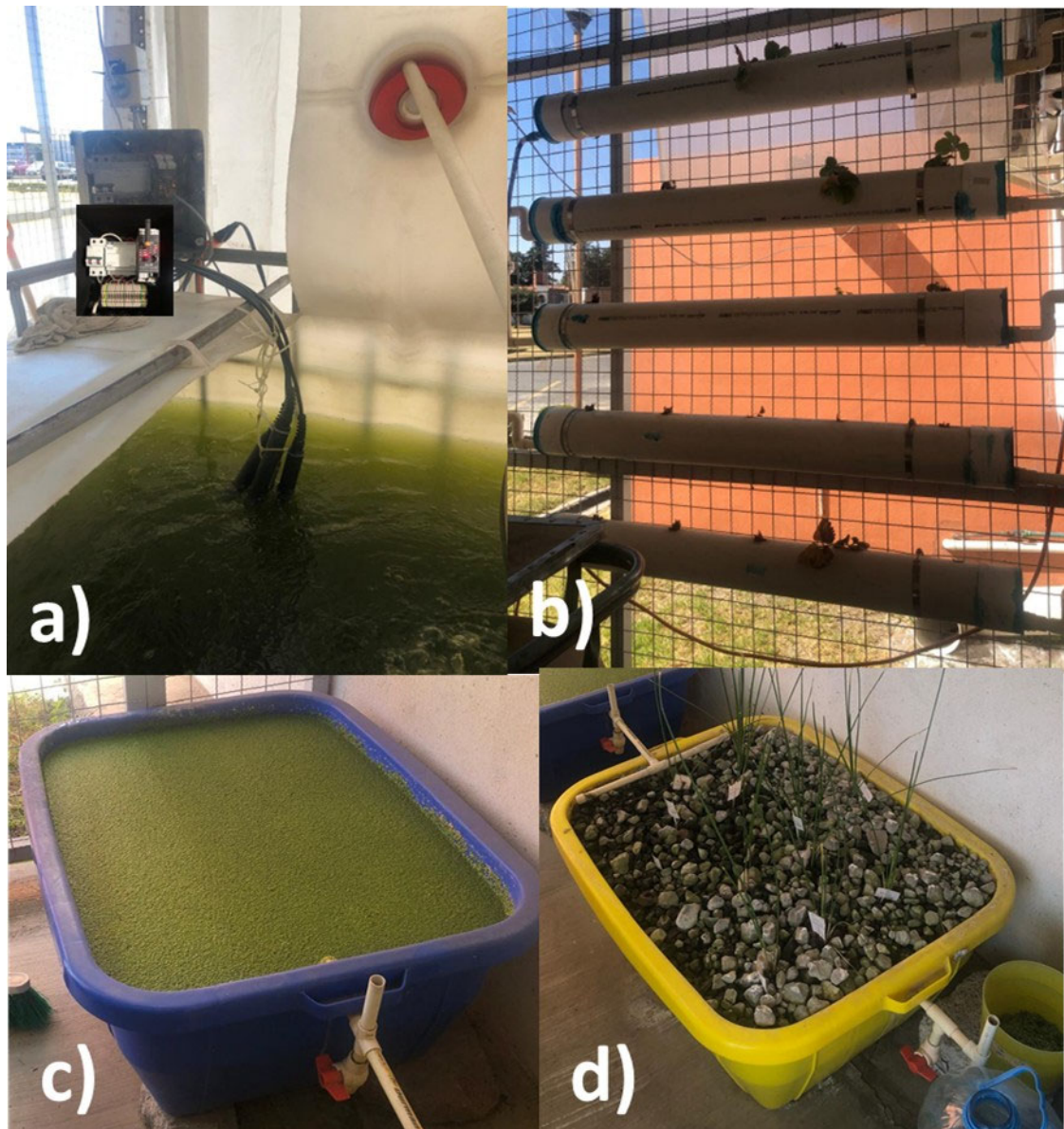


Figure 4. Aquaculture system used for real-time monitoring of water quality: (a) fish tank, (b) hydroponic strawberry cultivation, (c) *Lemna gibba* tank, and (d) tule wetland.

Figure 4b illustrates the hydroponic cultivation of strawberries within the system. Figure 4c shows the tank containing *Lemna gibba*, which functions as a biofilter and contributes to ammonia removal, achieving a reduction of approximately 92%, a value higher than the 71% reported for non-vegetated systems [16]. Finally, Figure 4d presents the constructed wetland planted with tule, which also acts as a biological filtration unit within the aquaponics system.

The monitoring system measured temperature, pH, oxidation–reduction potential (ORP), dissolved oxygen (DO), electrical conductivity, total dissolved solids (TDS), salinity,

turbidity, and total suspended solids (TSS) in the fish culture tank, generating a comprehensive dataset describing the behavior of these parameters throughout the monitoring period. The collected measurements were processed by the coordinator node, while the Real-Time Clock (RTC) module ensured precise time synchronization and accurate data logging.

The system was powered by a 12 V direct current (DC) adapter selected for its compatibility and efficiency in supplying both the coordinator node and the sensing node. The results demonstrate that the selected power supply enabled continuous and stable operation of the monitoring system throughout the entire monitoring period. No issues related to overheating or power instability were observed. Furthermore, the uninterrupted data transmission to the designated website confirms the reliable and sustained operation of the monitoring system.

The capability of the coordinating node to transmit data via email facilitated efficient integration with the IoT platform, ensuring that measured water-quality parameters were rapidly and accurately delivered to the designated monitoring interfaces. This functionality enables real-time data updates, supporting informed decision-making and improved management of environmental conditions within the aquaponics system [19].

3.4. Software

The Seal Graph Editor software v2.7.7 and the MODBUS RTU operating manual, which provides the configuration guidelines and communication parameters, were used to configure and program the inputs and outputs required for the correct operation of the Seneca datalogger model ZTE-WW. The system was programmed using modular architecture, which provides flexibility and scalability. This software structure consists of functional modules that can be assembled, modified, added, or removed independently without affecting the operation of other modules.

Communication between the datalogger and the sensors was configured with a transmission rate of 9600 baud, using 8 data bits, no parity, and 2 stop bits over a half-duplex RS-485 communication interface. During the monitoring period, the datalogger operated with an availability of 99.8% (6 failures per 10,000 readings), demonstrating its robustness under normal operating conditions in the aquaculture system.

3.5. Data Analysis

The complete real-time monitoring system was tested and validated over an approximately six-month period, from June to December 2024 (see Table 2). Statistical analysis of the data collected from the four monitoring nodes indicates that the system operated reliably, with 97% of the expected data successfully recorded during the monitoring period. This reliability exceeds the 95.2% data availability reported by [17] for comparable monitoring systems. The data obtained through monitoring software provides a detailed characterization of the physicochemical conditions of the water throughout the study period. The monitored parameters included temperature, dissolved oxygen, pH, and turbidity, which are key indicators for assessing water quality and its potential effects on aquatic ecosystems [29].

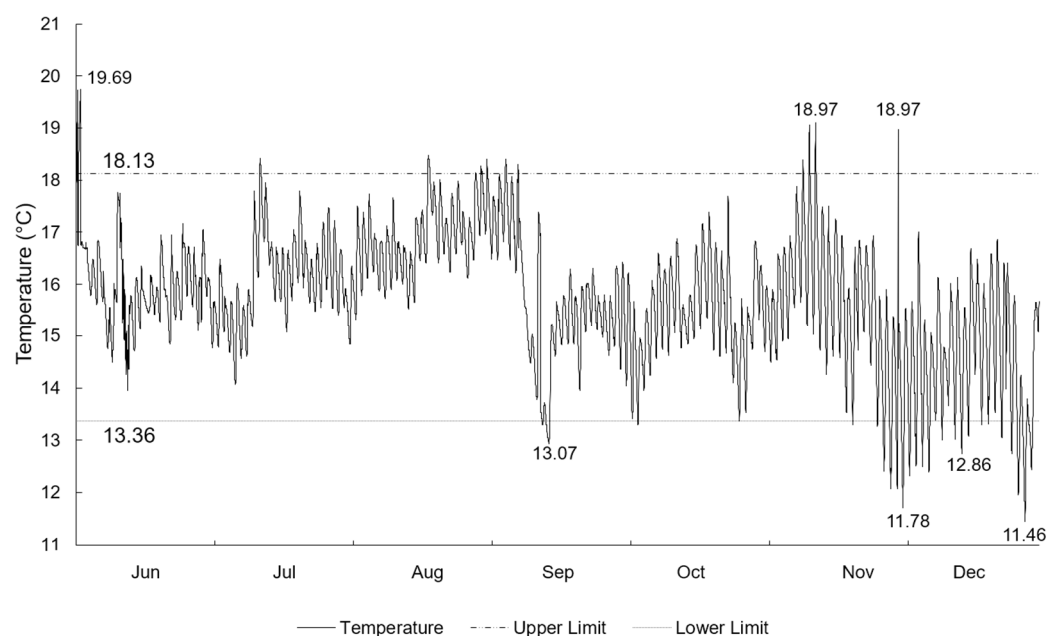
Table 2 summarizes the descriptive statistics of the monitored water-quality parameters. Temperature showed a mean value of 15.75 ± 1.19 °C, with minimum and maximum values of 11.46 °C and 19.69 °C, respectively, and a coefficient of variation (CV) of 7.6%. Electrical conductivity presented a mean value of 898 ± 89 µS/cm, ranging from 81 to 1264 µS/cm, with a CV of 10%. Salinity exhibited a mean value of 0.481 ± 0.048 ‰, with minimum and maximum values of 0.043 ‰ and 0.678 ‰, respectively, and a CV of 9.9%. Total dissolved solids (TDS) had a mean value of 453.3 ± 44.9 mg/L, with values ranging from 40.7 to 638 mg/L and a CV of 9.9%.

Table 2. Summary of the statistical data obtained for the different parameters measured in the water by the online monitoring system during the period of 6 months.

	Temperature (°C)	Electrical Conductivity (µS/cm)	Salinity (‰)	Total Dissolved Solids (mg/L)	Dissolved Oxygen (mg/L)	pH	Turbidity (NTU)	Total Suspended Solids (mg/L)
Mean	15.75	898	0.481	453.3	6.92	8.47	27.5	42.0
Median	15.78	912	0.489	460.2	6.58	8.50	24.4	37.6
Standard Deviation	1.19	89	0.048	44.9	2.27	0.36	14.9	21.5
Minimum data	11.46	107	0.043	40.7	1.34	6.42	3.7	5.3
Maximum data	19.69	1246	0.678	638.0	17.49	9.12	70.0	99.9
Coefficient of variation (%)	7.6	10	9.9	9.9	32.8	4.2	54.2	51.3

Dissolved oxygen (DO) showed a mean value of 6.92 ± 2.27 mg/L, with minimum and maximum values of 1.34 and 17.49 mg/L, respectively, and a CV of 32.8%. The pH values presented a mean of 8.47 ± 0.36 , ranging from 6.42 to 9.12, with a CV of 4.2%. Turbidity exhibited a mean value of 27.5 ± 14.9 NTU, with values between 3.7 and 70 NTU, resulting in a CV of 54.2%. Finally, total suspended solids (TSS) showed a mean value of 42 ± 21.5 mg/L, with minimum and maximum values of 5.3 and 99.9 mg/L, respectively, and a CV of 51.3%.

Figure 5 illustrates the variation in water temperature recorded in the fish tank by the real-time monitoring system during the six-month monitoring period. Water temperature ranged from a minimum of 11.46 °C in December to a maximum of 19.69 °C in June, with an average value of 15.75 ± 1.19 °C. This seasonal variation is expected and can significantly influence biological activity, metabolic rates of aquatic organisms, and nutrient dynamics in aquaculture systems. The average temperature observed in this study indicates moderate thermal conditions that may be suitable for a wide range of aquatic species [30].

**Figure 5.** Variation in temperature and outlier data obtained through real-time monitoring of the fish tank in the aquaponics system.

The temporal pattern of temperature follows a typical seasonal trend. A relatively warm and stable period was observed from June to September, characterized by moderate temperatures with limited fluctuations. This was followed by a gradual decline beginning in October, marking the transition to colder conditions typical of temperate climates. During

November and December, temperatures stabilized at lower values, confirming the onset of the winter season.

Figure 6 shows the variation in electrical conductivity recorded by the real-time monitoring system during the six-month monitoring period. The average electrical conductivity was $898 \pm 89 \mu\text{S}/\text{cm}$, with a minimum value of $107 \mu\text{S}/\text{cm}$ observed in December and a maximum value of $1246 \mu\text{S}/\text{cm}$ in September. Overall, electrical conductivity remained relatively stable throughout the monitored period, with most data points falling within the expected operational range and only a few values identified as outliers.

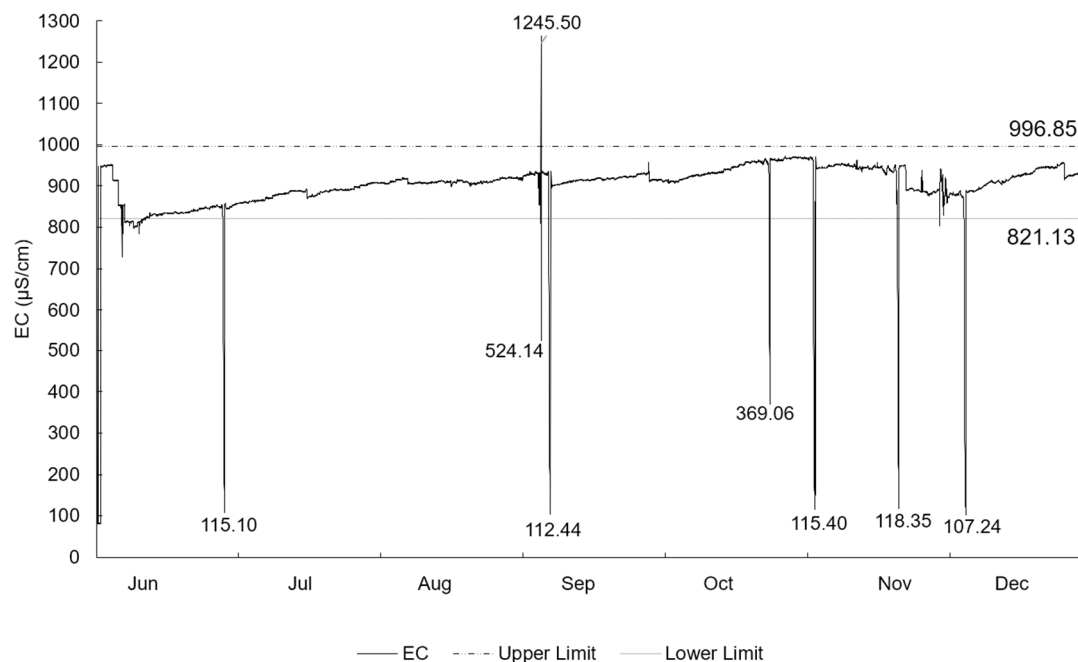


Figure 6. Variation in electrical conductivity and outlier data obtained through real-time monitoring of the fish tank in the aquaponics system.

This stability indicates that the concentration of dissolved ionic species and total dissolved solids remained relatively constant in the aquaponics system, which is beneficial for maintaining stable water chemistry and supporting fish health.

Figure 7 presents the variation in dissolved oxygen (DO) concentration measured in the fish tank during the monitoring period. The average DO concentration was $6.92 \pm 2.27 \text{ mg}/\text{L}$, with a minimum value of $1.34 \text{ mg}/\text{L}$ recorded in December and a maximum value of $17.49 \text{ mg}/\text{L}$ observed in November. These values indicate that the aquaculture system experienced significant fluctuations in oxygen availability during the monitoring period.

Such variability can be mainly attributed to the management of the aeration system. In the studied system, aeration was not continuous but was manually activated when operators detected DO concentrations approaching approximately $2 \text{ mg}/\text{L}$. In addition to aeration management, seasonal changes, biological activity, and physicochemical processes may also influence oxygen production and consumption dynamics in the fish tank.

The mean DO concentration of $6.92 \text{ mg}/\text{L}$ is generally suitable for the cultivation of common carp (*Cyprinus carpio*), a species known for its tolerance to moderate environmental fluctuations. However, the large variability observed suggests that the system operates under unstable oxygen conditions. This instability indicates that the aeration strategy is neither automated nor optimized, which may result in an imbalance between oxygen consumption and oxygen supply in the system [31].

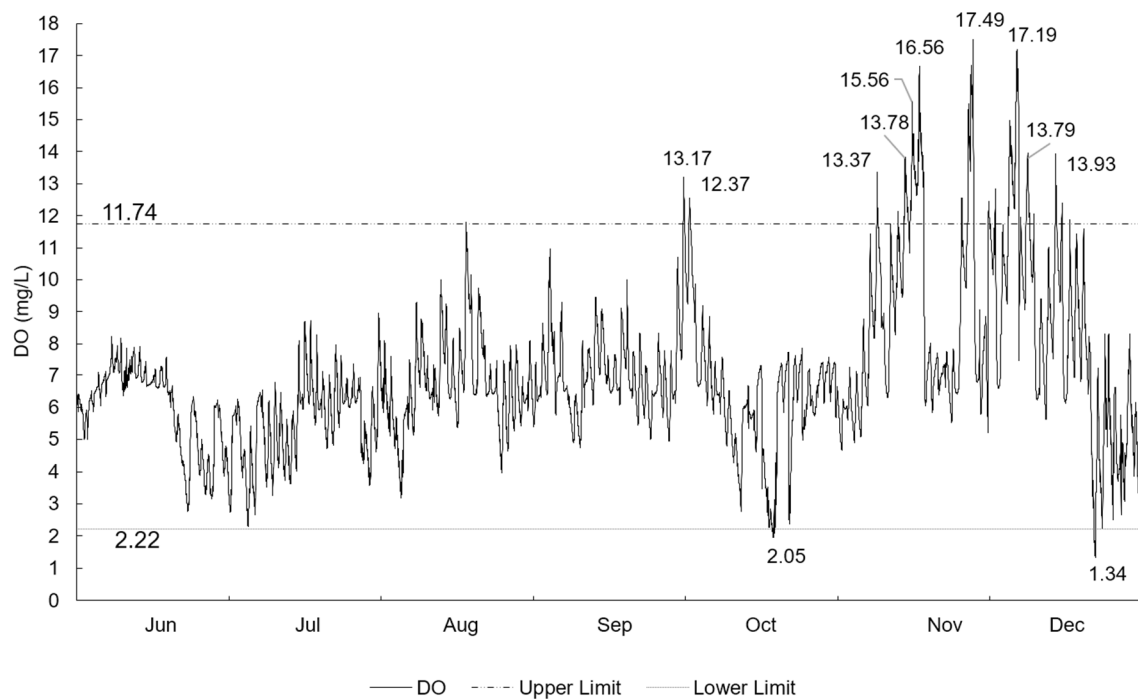


Figure 7. Variation in dissolved oxygen and the outlier data obtained through real-time monitoring of the fish tank in the aquaponics system.

It is noteworthy that unusually high DO values were observed during November and December, with concentrations ranging between 13 and 17.5 mg/L. These elevated values are likely associated with improper operation of the aeration system. Excessive dissolved oxygen levels in aquaculture systems are often related to total dissolved gas (TDG) supersaturation, a condition in which the combined partial pressures of dissolved gases exceed atmospheric pressure.

Exposure to supersaturated conditions can represent a physiological stress factor for fish and may lead to gas bubble disease (GBD). In carp species, clinical symptoms of GBD include abnormal swimming behavior, respiratory distress, exophthalmia, subcutaneous gas emboli, and damage to gill tissues [32]. Although *C. carpio* exhibits relatively higher tolerance to gas supersaturation compared to some pelagic fish species, prolonged exposure or higher TDG levels significantly increase the risk of morbidity and mortality.

Operational practices such as excessive oxygen injection, rapid temperature changes, or pressurized water inflows may unintentionally induce gas supersaturation. Effective management strategies include controlled aeration for degassing, avoidance of excessive oxygen input, and continuous monitoring of DO levels to maintain appropriate conditions for fish welfare. In the studied aquaponics system, the absence of automated DO control suggests that improvements in aeration management are necessary to ensure stable oxygen conditions and prevent potential stress to the cultured fish.

It is also important to note that some outlier values were primarily observed between 09:00 and 11:00 a.m., which coincides with the fish feeding period. These anomalies are likely associated with particle resuspension caused by fish activity and feed distribution near the sensors rather than failures in the monitoring system. Similar observations have been reported in IoT-based aquaculture monitoring systems [19].

Fortunately, these outliers represent isolated biological disturbances rather than actual deterioration in water quality. Nevertheless, the results highlight the importance of installing protective housings or shields around sensors located in zones of high turbulence, which can help reduce measurement interference and improve data stability.

Figure 8 shows the variation in pH values recorded by the real-time monitoring system during the monitoring period. The average pH of the water was 8.47 ± 0.36 , indicating slightly alkaline conditions, with a minimum value of 6.42 observed in June and a maximum value of 9.12 in December. The decrease in pH to values close to 6.42 in June suggests a temporary acidification process that may be associated with the accumulation of CO_2 resulting from fish and microbial respiration, the production of organic acids from the decomposition of organic matter (feces and uneaten feed), and nitrification processes that consume alkalinity [21].

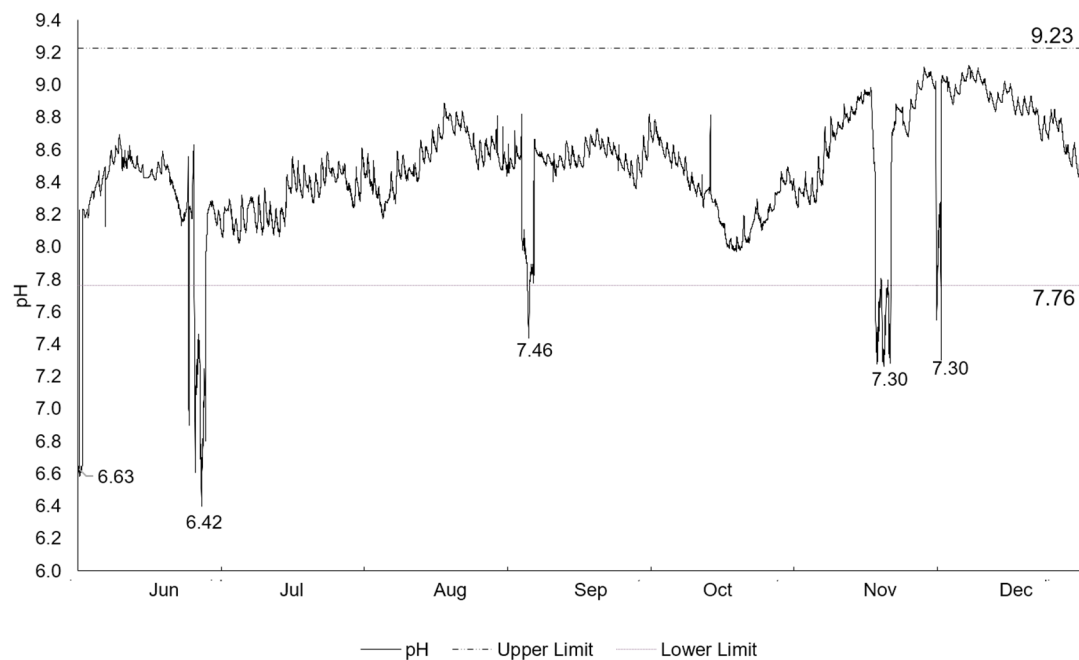


Figure 8. Variation in pH and outlier data obtained through real-time monitoring of the fish tank in the aquaponic system.

Overall, the pH values remained within the optimal range commonly recommended for aquaculture systems (6–9). Although the values of 6.42 and 6.63 recorded in June are close to the lower limit of this range, they remain within tolerable conditions for the species studied (*Cyprinus carpio*), though approaching critical thresholds for optimal physiological performance [33]. The observed pH fluctuations highlight the importance of continuous monitoring and proactive management of water chemistry in intensive aquaculture systems.

Figure 9 illustrates the variation in turbidity measured by the real-time monitoring system during the monitoring period. Turbidity values ranged from a minimum of 3.7 NTU to a maximum of 70 NTU, with an average value of 27.5 ± 14.9 NTU. This wide range indicates substantial fluctuations in the physical characteristics of the water, which may directly influence fish physiology and behavior.

The observed variability suggests episodic inputs of suspended particles due to processes such as nutrient inflow, sedimentation, and particle resuspension within the fish tank. These processes can reduce light penetration and potentially affect primary productivity within the system. Turbidity and total suspended solids (TSS) also exhibited the highest standard deviation and coefficient of variation among the monitored parameters. These fluctuations are likely related to uneaten feed, fish activity, and sediment disturbance near the sensors.

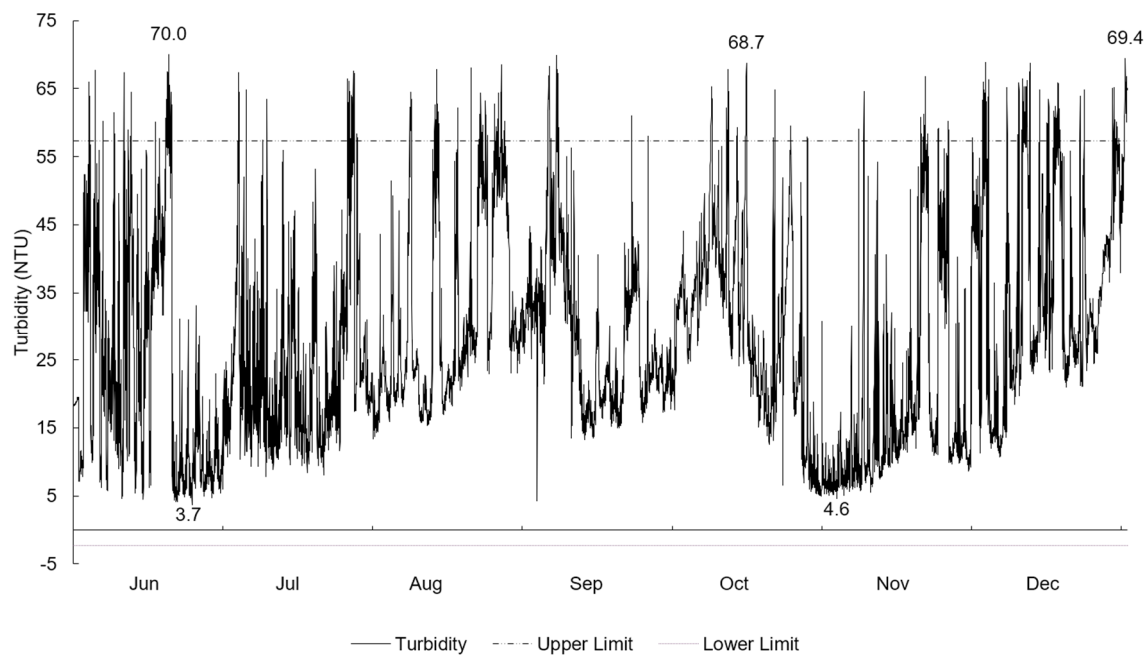


Figure 9. Variation in turbidity and outlier data obtained through real-time monitoring of the fish tank in the aquaponics system.

The variability observed in the water-quality parameters measured by the real-time monitoring system indicates that these fluctuations must be considered by system operators when managing aquaponics systems. Continuous monitoring allows operators to identify changes in physicochemical conditions that may affect fish production and overall system stability.

The results demonstrate that the aquaponics ecosystem is influenced by both internal factors (biological activity, feeding practices, and organic matter decomposition) and external factors (seasonal temperature changes and environmental conditions). These influences collectively regulate the physicochemical dynamics of the system during the monitoring period.

Real-time monitoring systems have become transformative tools in modern aquaponics and aquaculture by replacing traditional labor-intensive manual monitoring with automated and continuous data acquisition. The integration of advanced technologies such as Artificial Intelligence (AI), edge computing, and predictive analytics is expected to further enhance these systems by enabling fully autonomous and proactive management strategies.

Continuous monitoring provides significant advantages for aquaculture producers. Real-time data collection enables immediate detection of deviations in key parameters such as dissolved oxygen or pH, allowing corrective actions before biological stress occurs [34]. Systems integrated with automated controls can further optimize resource use, for example, by activating aeration only when dissolved oxygen levels fall below predefined thresholds [35]. Additionally, real-time monitoring supports the concept of precision aquaculture, where data-driven decision-making improves feed efficiency and reduces environmental impacts by minimizing waste generation [36].

Recent technological developments have also reduced the cost of implementing monitoring systems. The use of low-cost microcontrollers such as ESP8266 and Raspberry Pi platforms has significantly increased accessibility, with some systems costing substantially less than commercial monitoring solutions [35,37].

Despite their advantages, real-time monitoring systems still face several technical and economic challenges. One of the main technical limitations is sensor biofouling, which

involves the accumulation of microalgae and organic matter on sensor surfaces. This phenomenon can reduce measurement accuracy and shorten sensor lifespan [38]. Regular maintenance and recalibration are therefore necessary to prevent sensor drift and maintain measurement reliability [39].

Economic barriers may also limit adoption. While basic sensors for temperature and pH are relatively inexpensive, high-precision sensors for parameters such as dissolved oxygen and turbidity remain costly, which may restrict their use in low-budget aquaculture systems [29]. In addition, the dependence on reliable internet connectivity for cloud-based data management and the need for technical expertise among system operators represent additional barriers to widespread adoption [40].

A critical requirement for the adoption of real-time monitoring technologies is their ability to achieve accuracy levels comparable to conventional laboratory methods. Recent studies have reported strong correlations between sensor measurements and laboratory reference analyses. For example, an IoT-based monitoring system achieved accuracy levels of 96.51% for pH, 98.19% for total dissolved solids (TDS), and 97.03% for turbidity when compared with laboratory measurements and dissolved oxygen achieved a correlation coefficient of $R^2 = 0.999$ with standardized laboratory methods [4,18].

In aquaponic systems, real-time monitoring technologies have also demonstrated comparable performance to manual measurements for parameters such as temperature and pH, with some temperature sensors showing error rates below 7% [29,41]. These results indicate that modern sensor technologies can provide reliable measurements suitable for most commercial and research applications in aquaponics.

The future of water-quality monitoring lies in the integration of Artificial Intelligence (AI) and edge computing, moving beyond simple data collection toward intelligent prediction and automated management of aquaculture systems. Several emerging technological trends are shaping the next generation of monitoring platforms:

1. AI-driven predictive analytics: Machine learning models, such as Long Short-Term Memory (LSTM) neural networks, are increasingly used to predict water-quality trends with accuracies of up to 94.2%, enabling system managers to anticipate potential problems before they occur [42,43].
2. Edge computing and TinyML: By processing data locally on low-power devices, monitoring systems can significantly reduce latency and bandwidth requirements, improving operational reliability in remote or connectivity-limited areas [44,45].
3. Disease detection and computer vision: AI-based image analysis systems are increasingly capable of identifying fish diseases, detecting abnormal behavior, and estimating fish biomass in real time through integrated camera modules [46].
4. Hybrid AI models: The development of scalable and adaptive hybrid AI models will enable the optimization of complex aquaculture ecosystems by simultaneously considering environmental conditions, fish health, and system-level dynamics [47,48].

To assess the wider applicability of the proposed IoT platform beyond the *Cyprinus carpio* pilot setup [23], it is essential to highlight the operational flexibility of the sensing node integration. The core architecture relies on standard industrial communication protocols (Modbus RTU over RS-485), which means the datalogger can seamlessly adjust its polling intervals and alarm thresholds to match the tighter water quality boundaries required by more sensitive commercial species, such as rainbow trout (*Oncorhynchus mykiss*) or Nile tilapia (*Oreochromis niloticus*). For instance, shifting the operational parameters to a cold-water or high-salinity environment would only require adjusting software thresholds within the Seneca ZTE-WW datalogger framework without altering the physical hardware layout [44]. Regarding physical scale, the system's modular software architecture allows for the dynamic addition of multiple remote sensing nodes via the local RS-485 network

topology, enabling continuous validation in multi-tank commercial facilities or large-scale recirculating aquaculture systems (RAS) where high temporal data density is critical [41,49].

The claim of the system being ‘versatile and cost-effective’ is structurally substantiated by its hardware configuration and integration strategy. Unlike traditional commercial water quality stations that rely on closed-source, high-cost proprietary dataloggers and mandatory subscription cloud services, our system integrates robust, industrial-grade sensors (Aqualabo series) directly managed by a flexible, commercially accessible telemetry unit. Versatility is demonstrated by the multi-layered software design (connectivity, security, and scalability), which utilizes free automated workflows (power automate) and existing cloud storage infrastructure (Microsoft OneDrive), completely bypassing the recurring data-hosting fees typical of industrial monitoring [10,44]. Furthermore, by optimizing the acquisition routine to read data every 15 min and implementing an automated notification logic via standard SMTP, the setup drastically reduces manual labor hours for filtration verification and minimizes structural risks like dissolved gas supersaturation or extreme pH shifts through early alerts [12,50]. This strategic balance between highly accurate, factory-calibrated industrial probes and open, flexible data-handling protocols yields a total implementation cost significantly below equivalent commercial solutions, rendering precision aquaculture accessible for research and small-to-medium enterprise (SME) scale production.

In order to verify the practical applicability of the developed IoT platform, its cost efficiency must be evaluated across three operational dimensions:

Zero-Subscription Software Architecture: Unlike proprietary commercial stations that enforce recurring cloud-hosting fees, this platform pairs industrial-grade probes with the Seneca ZTE-WW gateway. By routing automated SMTP logs through free Power Automate workflows into Microsoft OneDrive, software-derived operational expenditures (OpEx) are entirely eliminated, successfully bypassing recurring commercial data-hosting fees [44].

Logistical and Labor Optimization: Traditional water monitoring relies heavily on manual sampling, requiring continuous labor, transport, and laboratory chemical consumables. Shifting the routine tracking to the automated, 15-min sensor acquisition protocol significantly reduces manual labor overhead and routine laboratory expenses over extended deployments, demonstrating efficiencies comparable to specialized edge-computing configurations deployed in integrated systems [12,44].

Bio-Economic Risk Mitigation: In intensive aquaponics, critical threshold breaches, such as sharp dissolved oxygen drops or nitrification-induced acidification, can cause catastrophic biomass mortality within hours. Conventional sampling twice a week introduces a dangerous operational blind spot. By dispatching immediate automated alerts via SMS and e-mail, the system enables rapid interventions, safeguarding the biological investment (*Cyprinus carpio* and crops) and preventing severe financial losses [49,50].

4. Conclusions

The real-time water-quality monitoring system implemented in the fish tank proved to be an effective tool for evaluating the behavior of key physicochemical parameters and validating the operational performance of the aquaponics system.

The monitoring system developed in this study demonstrated high versatility due to its modular programming structure, which allows easy modification and adaptation to different monitoring requirements. In addition, the system is economical and semi-portable, enabling reliable measurements even in environments where conventional water-quality monitoring systems may face operational limitations. The use of real-time sensors for water-quality monitoring can contribute to significant operational benefits by optimizing

resource use, reducing maintenance costs, and improving overall energy efficiency in aquaponics systems.

Continuous real-time monitoring generated a comprehensive database of water-quality parameters, minimizing sampling errors and eliminating potential contamination associated with sample transport, storage, and laboratory processing. The availability of real-time information also enables faster and more efficient decision-making when monitored parameters deviate from recommended operational ranges.

Statistical analysis showed no significant differences ($p > 0.05$) between the measurements obtained from the real-time monitoring system and those obtained through laboratory analyses for temperature, pH, electrical conductivity (EC), total dissolved solids (TDS), salinity, total suspended solids (TSS), and dissolved oxygen (DO). The system achieved data reliability of 97%, demonstrating its robustness and measurement accuracy under real operating conditions.

The developed monitoring system can be applied to both small-scale and larger aquaculture systems, provided that the monitored water quality remains within the operational range of the installed sensors. Although certain challenges remain, particularly regarding sensor maintenance and system implementation costs, these limitations can be mitigated through improved technologies and proper maintenance protocols.

Overall, real-time monitoring systems represent a transformative tool for aquaponics and aquaculture management by enabling continuous supervision of water-quality conditions. As emerging technologies such as Artificial Intelligence and edge computing become integrated into these systems, future monitoring platforms will likely provide predictive capabilities that further enhance system stability, reduce operational risks, and improve production efficiency.

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Institutional Review Board Statement: Ethical review and approval were waived for this study due to the strictly non-invasive, passive nature of the research, which focused exclusively on engineering, electronic telemetry, and environmental data acquisition. The methodology relied entirely on submerged, passive industrial sensors to continuously monitor the water matrix's physicochemical parameters without handling, manipulating, or altering the biological processes of the aquatic specimens. Furthermore, under current Mexican federal legislation, the official standard for laboratory animals (NOM-062-ZOO-1999) excludes fish from its regulatory scope, while the General Law of Sustainable Fisheries and Aquaculture (Ley General de Pesca y Acuicultura Sustentables) mandates formal ethical oversight or approval codes strictly for active biological interventions, toxicological bioassays, invasive tracking, or animal sacrifice. Consequently, since this study did not perform any regulated biological interventions and the fish biomass belonged to a separate, pre-established pilot culture framework, a formal ethical approval code was legally waived.

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors upon request.

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